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Risa 3D

Analysis of a Stable 3D Bridge Structure

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Procedure:

Create a stable 3D bridge structure in Risa 3D. Compare the analysis of the structure with my handwritten calculations.

Results:

The following model and tables are the results of my structure after following all steps given in my lab handout. By viewing these tables you can determine the specific loading, and deflection at each point due to a moving loading (H20 specified in lab handout) throughout the structure from one joint to the next.

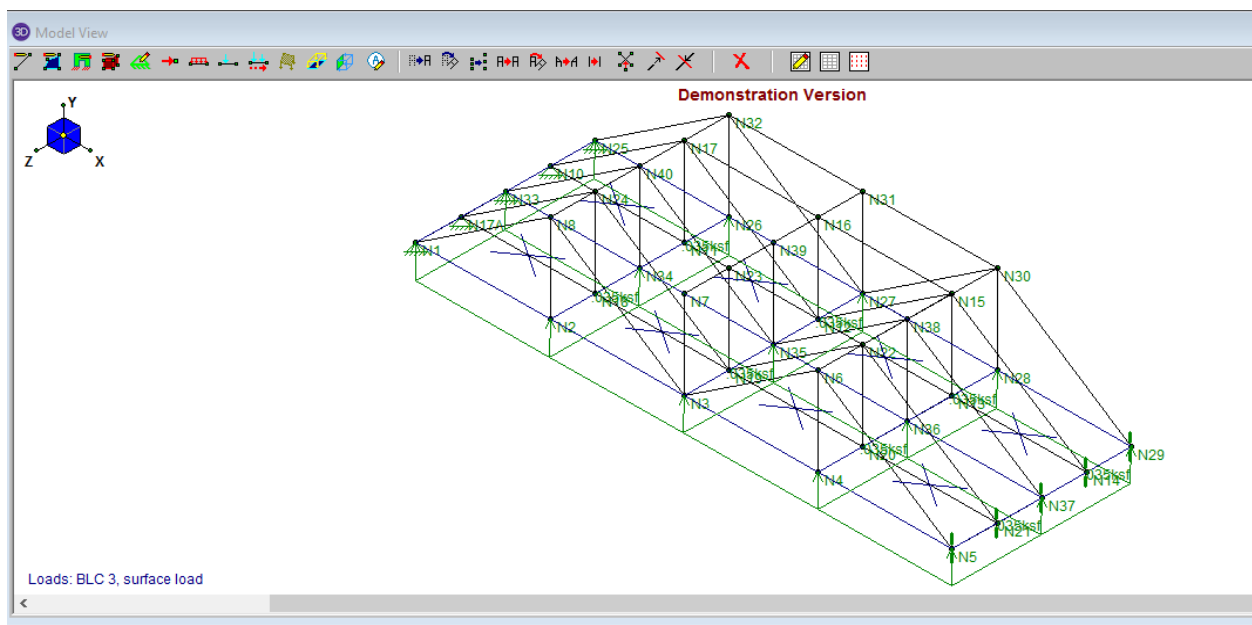


Figure 1: Structure created using Risa 3D

	L...	Joint Label	X [k]	Y [k]	Z [k]	MX [k-ft]	MY [k-ft]	MZ [k-ft]
1	1	N1	3.791	28.646	4.517	0	0	0
2	1	N5	0	28.649	0	0	0	0
3	1	N10	.19	10.11	0	0	0	0
4	1	N14	0	10.226	0	0	0	0
5	1	N17A	.193	5.248	0	0	0	0
6	1	N21	0	5.243	0	0	0	0
7	1	N25	3.792	28.569	-4.516	0	0	0
8	1	N29	0	28.65	0	0	0	0
9	1	N33	-7.966	42.473	-.001	0	0	0
10	1	N37	0	42.461	0	0	0	0
11	1	Totals:	0	230.273	0			
12	1	COG (ft):	X: 30.024	Y: 1.307	Z: -10.21			

Table 1: Shows Joint Reactions by combination due to a moving load being applied at each joint.

Demonstration - [C:\Users\Jasmine\Documents\RISADemo\Model Files\Backups\Lab 7_backup.r3d] - [Joint Deflection]

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1	N1	0	0	0	-2.554e-6	3.834e-5	-8.955e-4
1	N2	.004	-.161	0	-8.84e-4	-4.095e-6	-9.012e-4
1	N3	.009	-.243	0	-1.273e-3	-1.13e-5	-1.688e-6
1	N4	.013	-.162	0	-8.745e-4	8.777e-5	9.008e-4
1	N5	.018	0	0	-1.129e-6	-3.581e-4	8.996e-4
1	N6	-.03	-.15	-.003	4.037e-4	-5.059e-5	6.548e-4
1	N7	.009	-.244	0	6.154e-4	-7.768e-6	-1.227e-6
1	N8	.048	-.149	-.001	3.955e-4	1.081e-4	-6.557e-4
1	N10	0	0	0	1.335e-5	-3.658e-7	-6.63e-4
1	N11	.011	-.059	0	1.62e-5	-9.967e-7	-1.474e-4
1	N12	.022	-.077	0	1.285e-5	7.095e-7	0
1	N13	.033	-.059	0	2.327e-6	9.922e-6	1.473e-4
1	N14	.044	0	0	1.692e-6	-4.825e-5	6.632e-4
1	N15	.015	-.059	0	7.591e-6	6.749e-6	1.959e-4
1	N16	.022	-.079	0	5.602e-5	3.217e-6	0
1	N17	.03	-.059	.001	1.442e-5	3.91e-6	-1.959e-4
1	N17A	0	0	0	1.676e-7	-6.623e-6	-6.635e-4
1	N18	.011	-.06	0	-7.941e-6	1.957e-6	-1.478e-4
1	N19	.022	-.077	0	-2.044e-5	4.19e-6	-1.604e-8
1	N20	.033	-.06	0	-1.727e-5	-1.387e-5	1.478e-4
1	N21	.044	0	0	-1.295e-6	6.443e-5	6.635e-4
1	N22	.014	-.059	-.001	-2.529e-5	-5.353e-6	1.966e-4
1	N23	.022	-.079	0	-1.059e-4	8.016e-6	-1.894e-8
1	N24	.029	-.059				

Tables 2 and 3: shows the deflection at each joint due to the moving load (This is table 2)

		L...	Joint Label	X [in]	Y [in]	Z [in]	X Rotat...	Y Rotat...	Z Rotat...	
13		1	N14	.044	0	0	1.692e-6	-4.825e-5	6.632e-4	
14		1	N15	.015	-.059	0	7.591e-6	6.749e-6	1.959e-4	
15		1	N16	.022	-.079	0	5.602e-5	3.217e-6	0	
16		1	N17	.03	-.059	.001	1.442e-5	3.91e-6	-1.959e-4	
17		1	N17A	0	0	0	1.676e-7	-6.623e-6	-6.635e-4	
18		1	N18	.011	-.06	0	-7.941e-6	1.957e-6	-1.478e-4	
19		1	N19	.022	-.077	0	-2.044e-5	4.19e-6	-1.604e-8	
20		1	N20	.033	-.06	0	-1.727e-5	-1.387e-5	1.478e-4	
21		1	N21	.044	0	0	-1.295e-6	6.443e-5	6.635e-4	
22		1	N22	.014	-.059	-.001	-2.529e-5	-5.353e-6	1.966e-4	
23		1	N23	.022	-.079	0	-1.059e-4	8.016e-6	-1.894e-8	
24		1	N24	.029	-.059	0	-4.03e-6	2.087e-7	-1.966e-4	
25		1	N25	0	0	0	-2.063e-7	-7.728e-5	-8.955e-4	
26		1	N26	.004	-.161	0	8.85e-4	1.325e-5	-9.01e-4	
27		1	N27	.009	-.243	0	1.273e-3	1.326e-5	-1.811e-6	
28		1	N28	.013	-.162	0	8.744e-4	-8.713e-5	9.008e-4	
29		1	N29	.018	0	0	1.088e-6	3.538e-4	8.996e-4	
30		1	N30	-.03	-.15	.001	-4.198e-4	5.524e-5	6.548e-4	
31		1	N31	.009	-.244	0	-6.121e-4	1.702e-5	-1.241e-6	
32		1	N32	.048	-.149	.003	-3.796e-4	-9.995e-5	-6.557e-4	
33		1	N33	0	0	0	1.345e-6	1.448e-7	-1.218e-3	
34		1	N34	.004	-.215	0	-4.203e-7	-3.925e-7	-1.165e-3	
35		1	N35	.009	-.32	0	8.378e-8	1.486e-6	1.317e-6	
36		1	N36	.013	-.214	0	4.965e-8	1.002e-6	1.165e-3	
37		1	N37	.017	0	0	2.012e-8	-5.615e-6	1.214e-3	
38		1	N38	-.044	-.197	0	-7.653e-6	2.628e-6	8.363e-4	
39		1	N39	.009	-.321	0	2.193e-6	5.182e-6	1.068e-6	
40		1	N40	.061	-.197	0	1.01e-5	1.415e-6	-8.357e-4	

Tables 2 and 3: shows the deflection at each joint due to the moving load (This is table 3)

Conclusion:

After comparing my handwritten calculations with Risa 3D I was able to determine that the structure that I created in RISA 3D is sufficient to support the moving load that is being applied to it. Therefore, the structure is stable and all calculations are correct.